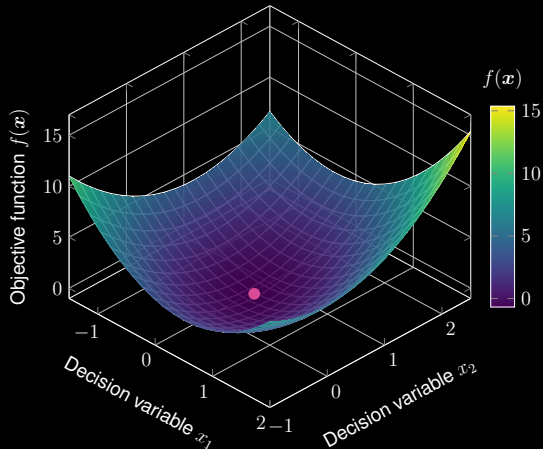
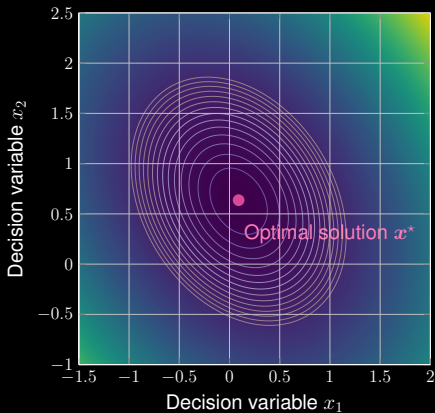


Quadratic Optimization

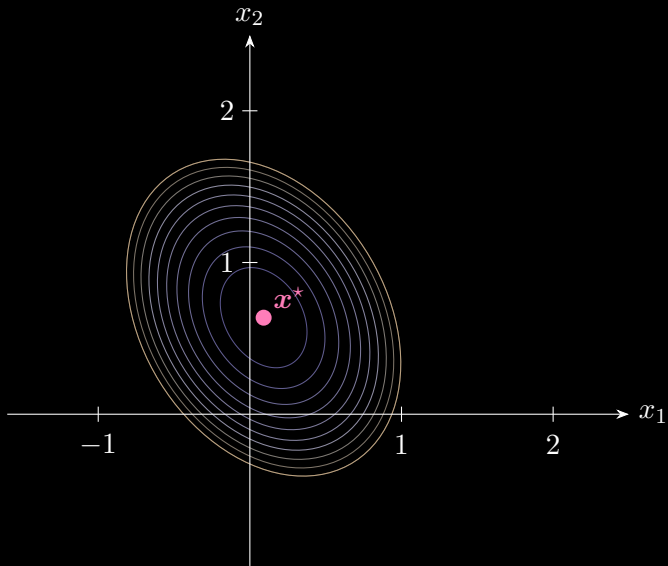
$$\min_{\mathbf{x} \in \mathbb{R}^2} \frac{1}{2} \mathbf{x}^\top \mathbf{A} \mathbf{x} - \mathbf{b}^\top \mathbf{x} \quad \text{with} \quad \mathbf{A} = \begin{bmatrix} 4 & 1 \\ 1 & 3 \end{bmatrix} \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$



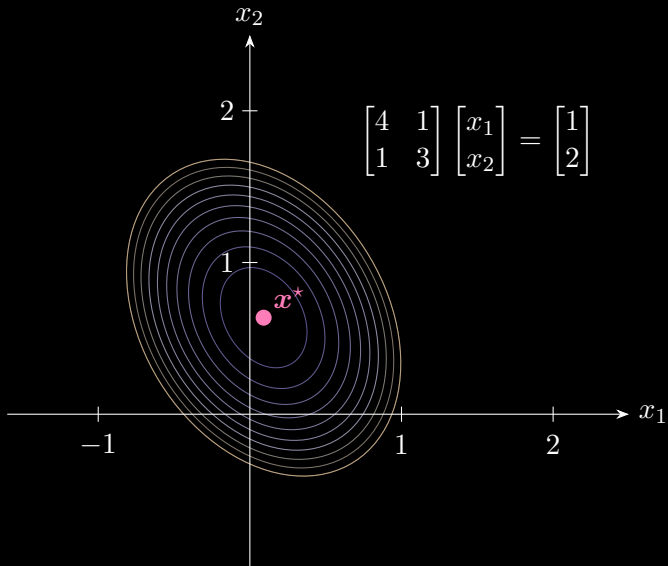
Quadratic Optimization

$$\min_{\mathbf{x} \in \mathbb{R}^2} \underbrace{\frac{1}{2} \mathbf{x}^\top \mathbf{A} \mathbf{x} - \mathbf{b}^\top \mathbf{x}}_{\text{Objective function } f} \quad \Rightarrow \quad \underbrace{\frac{d f}{d \mathbf{x}} = \mathbf{A} \mathbf{x} - \mathbf{b}}_{\text{First-order derivative}} = \mathbf{0} \quad \Rightarrow \quad \underbrace{\mathbf{A} \mathbf{x} = \mathbf{b}}_{\text{Linear system}}$$

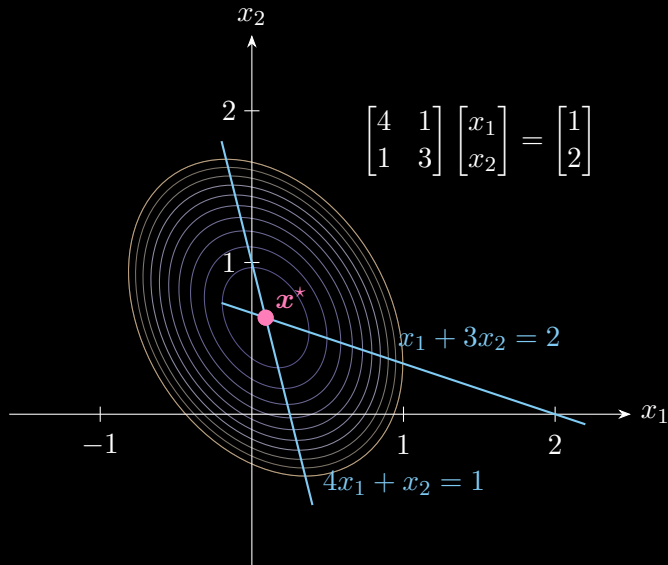
Quadratic Optimization



Equivalent Linear System $Ax = b$



Equivalent Linear System $Ax = b$



Update Equations

Essential idea: Conjugate gradient solves linear system $\mathbf{Ax} = \mathbf{b}$ iteratively.

❶ $\mathbf{x}$ is updated by

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \underbrace{\alpha_k}_{\text{step size (unknown)}} \mathbf{d}_k$$

with initial point $\mathbf{x}_0$.

❷ $\mathbf{d}_k$ is the search direction:

$$\mathbf{d}_{k+1} = \underbrace{\mathbf{r}_{k+1}}_{\text{residual}} + \underbrace{\beta_k}_{\text{step size (unknown)}} \mathbf{d}_k$$

③ How to write down the update of residual $\mathbf{r}_{k+1}$?

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \alpha_k \mathbf{d}_k$$

$$\Rightarrow \mathbf{Ax}_{k+1} = \mathbf{Ax}_k + \alpha_k \mathbf{Ad}_k$$

$$\Rightarrow \mathbf{b} - \mathbf{Ax}_{k+1} = \mathbf{b} - \mathbf{Ax}_k - \alpha_k \mathbf{Ad}_k$$

③ How to write down the update of residual $\mathbf{r}_{k+1}$?

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \alpha_k \mathbf{d}_k$$

$$\Rightarrow \mathbf{A}\mathbf{x}_{k+1} = \mathbf{A}\mathbf{x}_k + \alpha_k \mathbf{A}\mathbf{d}_k$$

$$\Rightarrow \mathbf{b} - \mathbf{A}\mathbf{x}_{k+1} = \mathbf{b} - \mathbf{A}\mathbf{x}_k - \alpha_k \mathbf{A}\mathbf{d}_k$$

$$\Rightarrow \mathbf{r}_{k+1} = \mathbf{r}_k - \alpha_k \mathbf{A}\mathbf{d}_k$$

To summarize, we have the following update equations to solve $\mathbf{Ax} = \mathbf{b}$

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \alpha_k \mathbf{d}_k \quad (\text{decision variables})$$

$$\mathbf{r}_{k+1} = \mathbf{r}_k - \alpha_k \mathbf{A} \mathbf{d}_k \quad (\text{residuals})$$

$$\mathbf{d}_{k+1} = \mathbf{r}_{k+1} + \beta_k \mathbf{d}_k \quad (\text{search direction})$$

△ But how to write down the update equations of α_k and β_k ?

Orthogonal residuals + Conjugate search directions ($\mathbf{A}$ -orthogonality)

$$\mathbf{r}_{k+1}^\top \mathbf{r}_k = 0 \quad \text{and} \quad \mathbf{d}_{k+1}^\top \mathbf{A} \mathbf{d}_k = 0$$

for any iterations k and $k + 1$.

④ Orthogonal Residuals $\mathbf{r}_k^\top \mathbf{r}_{k+1} = 0$:

$$\mathbf{r}_k^\top \underbrace{(\mathbf{r}_k - \alpha_k \mathbf{A} \mathbf{d}_k)}_{=\mathbf{r}_{k+1}} = 0 \quad \Rightarrow \quad \alpha_k = \frac{\mathbf{r}_k^\top \mathbf{r}_k}{\mathbf{r}_k^\top \mathbf{A} \mathbf{d}_k}$$

④ Orthogonal Residuals $\mathbf{r}_k^\top \mathbf{r}_{k+1} = 0$:

$$\mathbf{r}_k^\top \underbrace{(\mathbf{r}_k - \alpha_k \mathbf{A} \mathbf{d}_k)}_{=\mathbf{r}_{k+1}} = 0 \quad \Rightarrow \quad \alpha_k = \frac{\mathbf{r}_k^\top \mathbf{r}_k}{\mathbf{r}_k^\top \mathbf{A} \mathbf{d}_k}$$

○ Denominator:

$$\underbrace{\mathbf{r}_k^\top \mathbf{A} \mathbf{d}_k}_{\text{derived from the update equation of search direction}} = (\mathbf{d}_k - \beta_{k-1} \mathbf{d}_{k-1})^\top \mathbf{A} \mathbf{d}_k = \mathbf{d}_k^\top \mathbf{A} \mathbf{d}_k - \beta_{k-1} \underbrace{\mathbf{d}_{k-1}^\top \mathbf{A} \mathbf{d}_k}_{=0 \text{ (conjugate)}} = \mathbf{d}_k^\top \mathbf{A} \mathbf{d}_k$$

○ Update equation of step size α_k :

$$\alpha_k = \frac{\mathbf{r}_k^\top \mathbf{r}_k}{\mathbf{d}_k^\top \mathbf{A} \mathbf{d}_k}$$

⑤ Conjugate search directions $\mathbf{d}_{k+1}^\top \mathbf{A} \mathbf{d}_k = 0$:

$$\underbrace{(\mathbf{r}_{k+1} + \beta_k \mathbf{d}_k)}_{\mathbf{d}_{k+1}}^\top \mathbf{A} \mathbf{d}_k = 0 \quad \Rightarrow \quad \beta_k = -\frac{\mathbf{r}_{k+1}^\top \mathbf{A} \mathbf{d}_k}{\mathbf{d}_k^\top \mathbf{A} \mathbf{d}_k}$$

○ Considering

$$\mathbf{A} \mathbf{d}_k = \frac{\mathbf{r}_k - \mathbf{r}_{k+1}}{\alpha_k} \quad \alpha_k = \frac{\mathbf{r}_k^\top \mathbf{r}_k}{\mathbf{d}_k^\top \mathbf{A} \mathbf{d}_k}$$

○ Update equation of step size β_k :

$$\beta_k = \frac{\mathbf{r}_{k+1}^\top \mathbf{r}_{k+1}}{\mathbf{r}_k^\top \mathbf{r}_k}$$

Conjugate Gradient Method

To summarize, we have the following update equations to solve $\mathbf{Ax} = \mathbf{b}$

$$\alpha_k = \frac{\mathbf{r}_k^\top \mathbf{r}_k}{\mathbf{d}_k^\top \mathbf{A} \mathbf{d}_k} \quad (\text{step size})$$

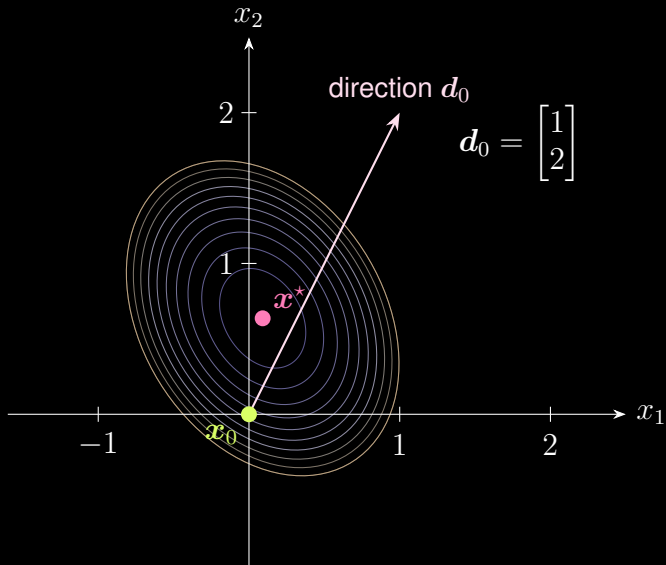
$$\mathbf{x}_{k+1} = \mathbf{x}_k + \alpha_k \mathbf{d}_k \quad (\text{decision variables})$$

$$\mathbf{r}_{k+1} = \mathbf{r}_k - \alpha_k \mathbf{A} \mathbf{d}_k \quad (\text{residuals})$$

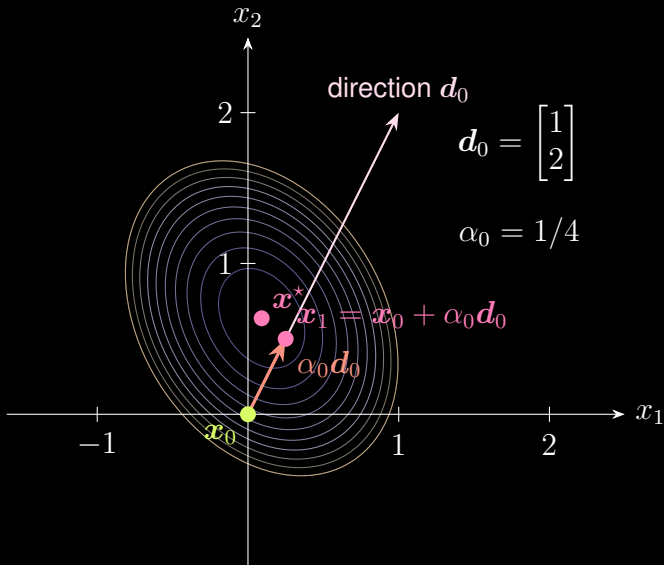
$$\beta_k = \frac{\mathbf{r}_{k+1}^\top \mathbf{r}_{k+1}}{\mathbf{r}_k^\top \mathbf{r}_k} \quad (\text{step size})$$

$$\mathbf{d}_{k+1} = \mathbf{r}_{k+1} + \beta_k \mathbf{d}_k \quad (\text{search direction})$$

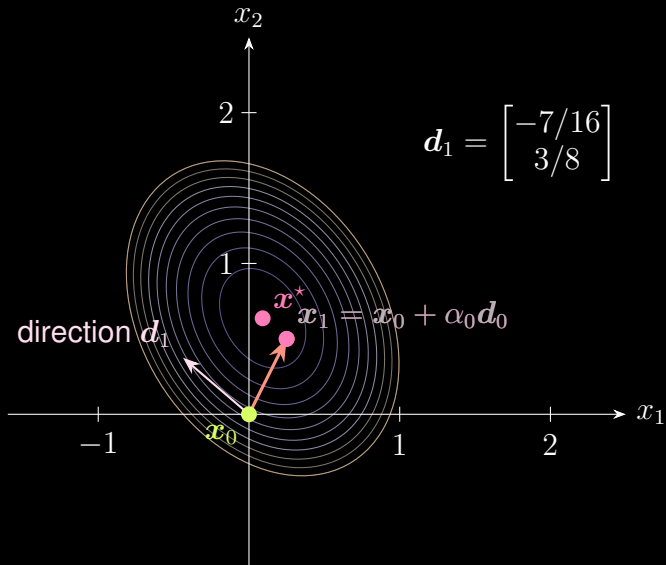
Conjugate Gradient Method



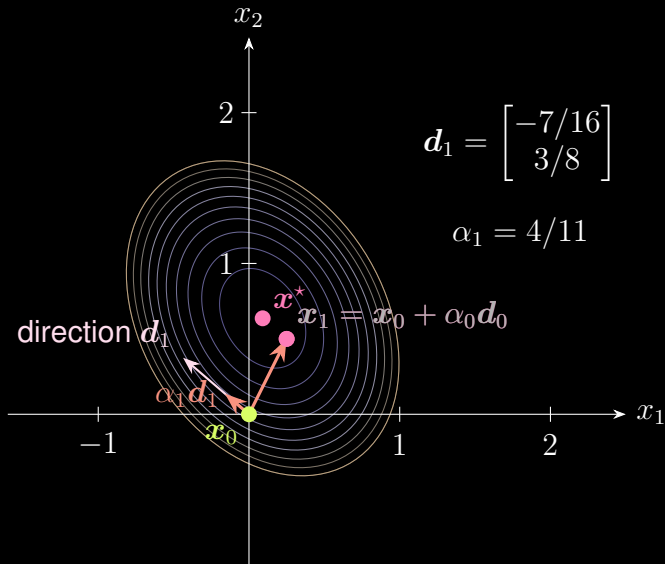
Conjugate Gradient Method



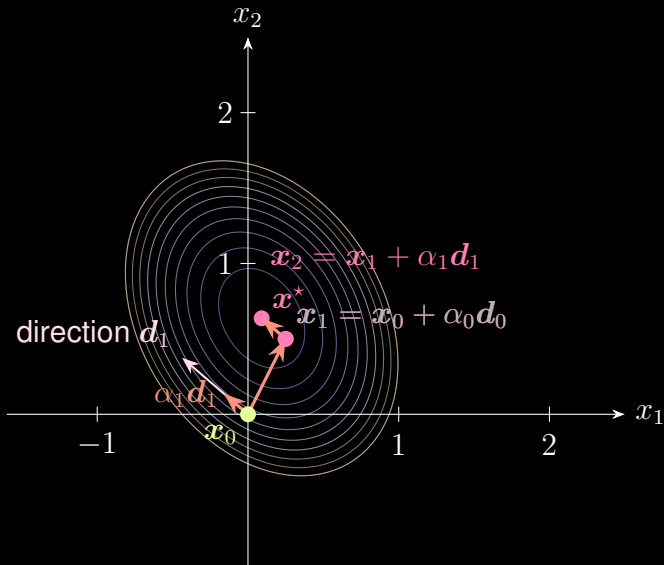
Conjugate Gradient Method



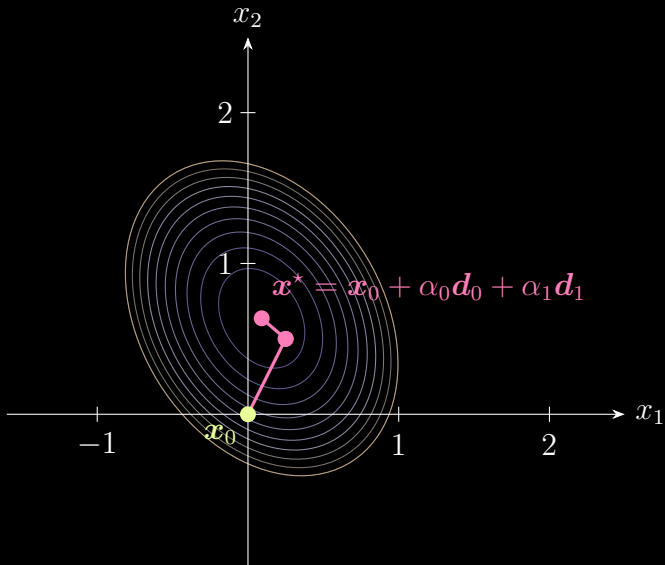
Conjugate Gradient Method



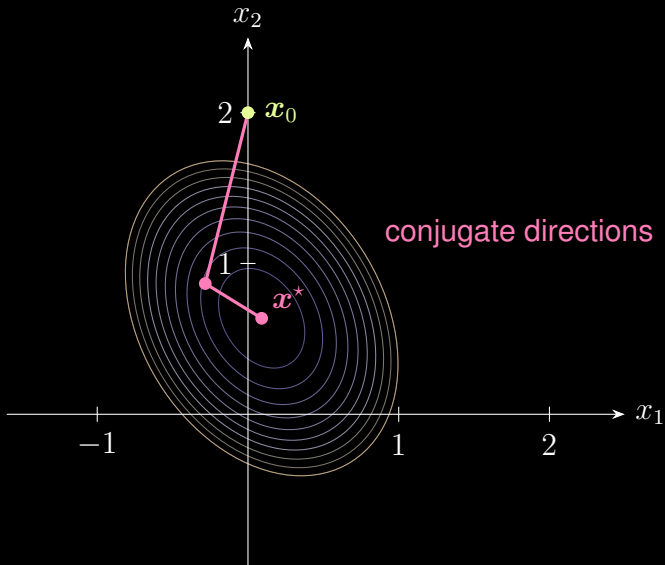
Conjugate Gradient Method



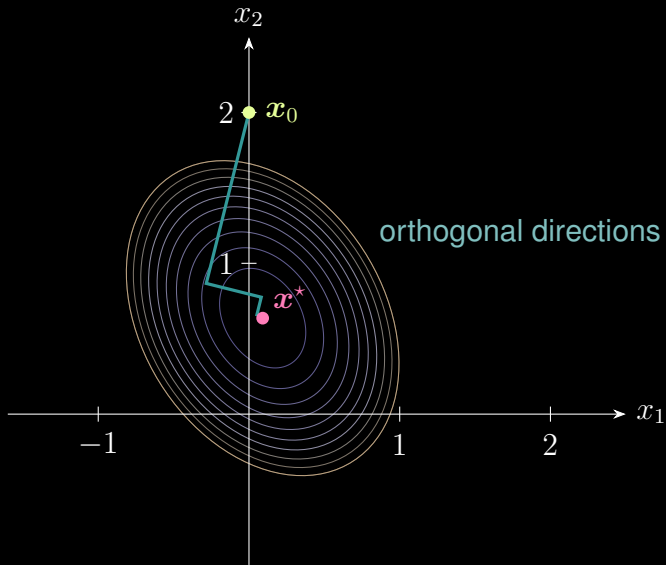
Conjugate Gradient Method



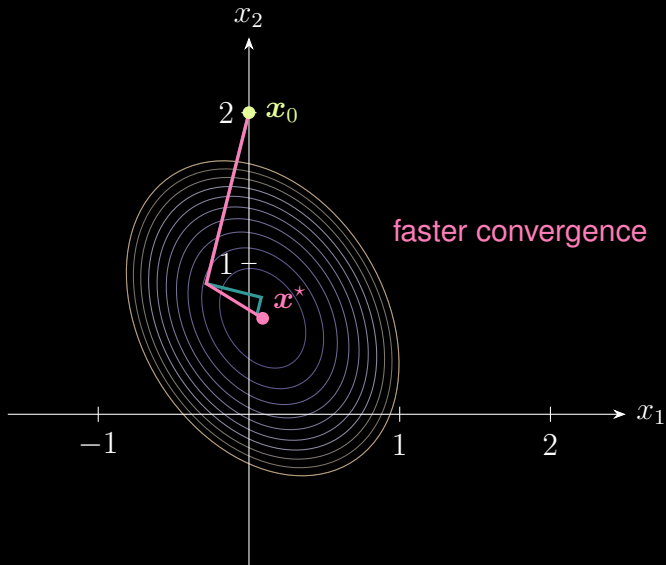
Conjugate Gradient Method



Steepest Gradient Descent Method

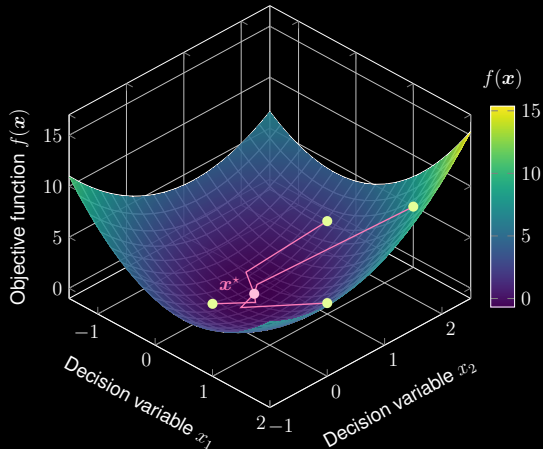
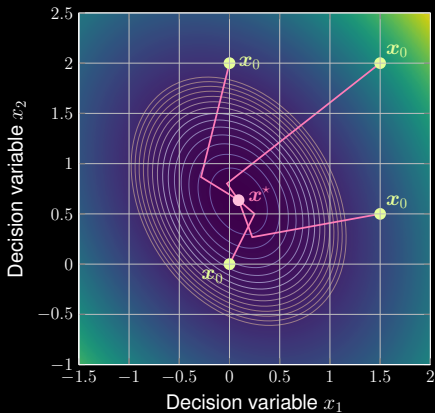


Conjugate Directions vs. Orthogonal Directions



Conjugate Gradient Method for Solving Quadratic Optimization

$$\min_{\mathbf{x} \in \mathbb{R}^2} \frac{1}{2} \mathbf{x}^\top \mathbf{A} \mathbf{x} - \mathbf{b}^\top \mathbf{x} \quad \text{with} \quad \mathbf{A} = \begin{bmatrix} 4 & 1 \\ 1 & 3 \end{bmatrix} \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$



Thanks for your attention!

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